Examiner only

7. (a) Carys and Pavel prepared crystals of zinc sulfate by reacting zinc carbonate with an acid.

(i) Give the name of the acid they used. [1]

....

(ii) In the first stage of their preparation, they added excess zinc carbonate to the acid.

I. Give the observation that **immediately** shows a reaction is taking place. [1]

....

II. State why they added **excess** zinc carbonate. [1]

....

(iii) Describe the remaining two stages they carried out to obtain a **pure** sample of zinc sulfate crystals. [2]

....

....

....

(iv) Give the chemical formula of zinc sulfate. [1]

....

(b) In another experiment, Carys and Pavel investigated the temperature rise when dilute hydrochloric acid neutralises sodium hydroxide solution.

The acid was added 5 cm^3 at a time to 25 cm^3 of sodium hydroxide solution. They recorded the highest temperature reached after each addition using a digital thermometer.

They obtained the following results. The result for 15 cm^3 of acid is missing.

Volume of acid added (cm^3)	Temperature (°C)
0	20.4
5	21.9
10	23.4
15	
20	26.4
25	26.3
30	25.9
35	25.5
40	25.1

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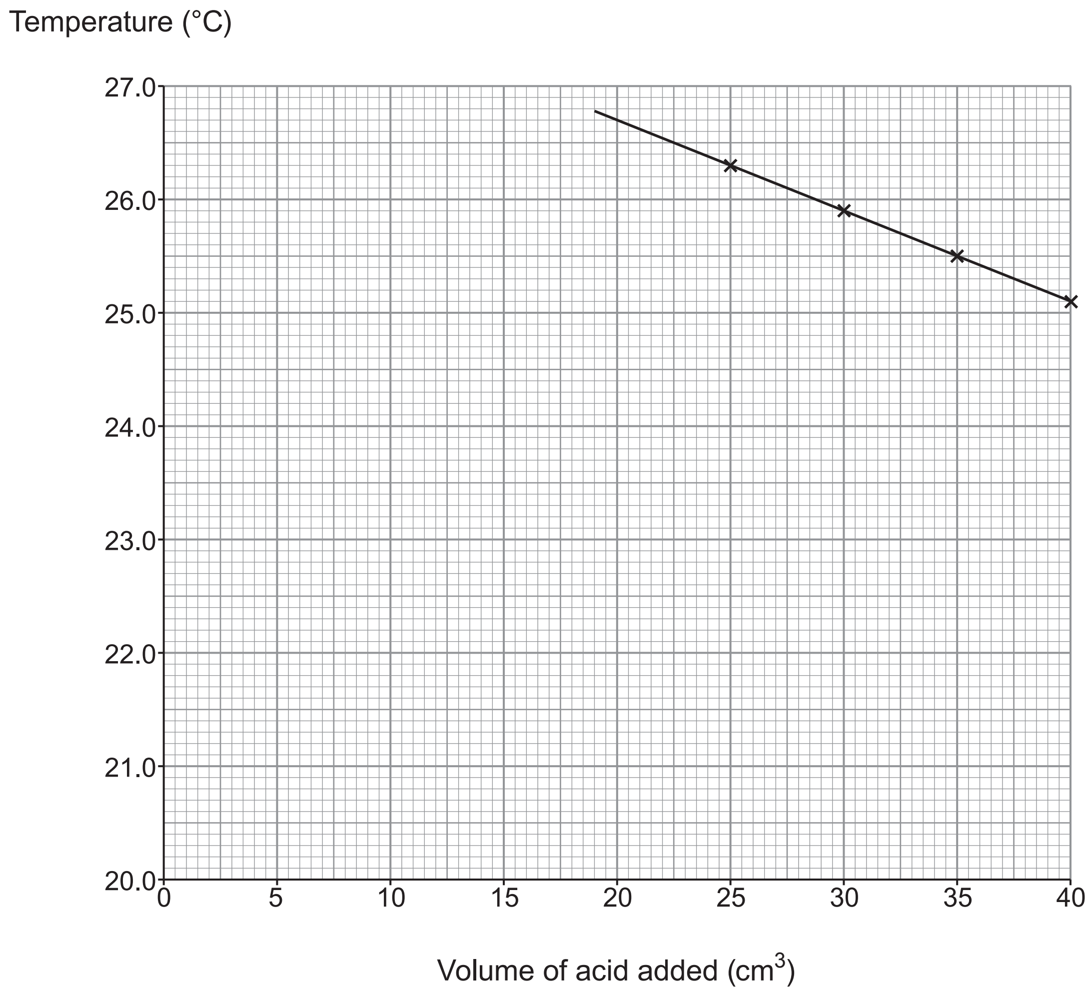
(i) The last four results have been plotted on the grid below and a straight line drawn through the points.

Plot the remaining four points and draw a **straight line** through them so that it intersects the line already drawn. [3]



(ii) I. Use your graph to give the temperature that would have been recorded when 15 cm^3 of acid was added. [1]

.... °C

II. Carys and Pavel concluded that the volume of acid needed to just neutralise all the sodium hydroxide solution was somewhere between 20 cm^3 and 25 cm^3.

Use the graph to suggest the exact volume of acid needed. [1]

.... cm^3

 (3430U50-1) **Turn over.**

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(iii) The temperatures recorded are slightly **lower** than expected.

Tick (✓) to show which **two** improvements to the method would enable Carys and Pavel to obtain results closer to the expected values. [2]

use a beaker instead of a flask	☐
repeat the method	☐
add the acid in smaller intervals	☐
wrap cotton wool around the flask	☐
use a larger flask	☐
place a lid on the flask	☐

(c) In a different experiment, it was found that the maximum temperature was reached when 40 cm^3 of hydrochloric acid was added to 20 cm^3 of sodium hydroxide solution.

State what this means in terms of the relative concentrations of the solutions used. [2]

....

....

....

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END OF PAPER

